

□ 04 Probability models – the Poisson distribution

Problem 1

Customers arrive to join a queue for an ATM at an average rate of $1\frac{1}{2}$ per minute but independent of other arrivals. How likely is it that (exactly) one customer will arrive in the next $2\frac{1}{2}$ minutes?

Problem 2

Potholes are to be found on a stretch of country back road with a pothole on average every 750 m. Assuming that the potholes occur at random what is the probability that a particular stretch of roadway of length 1,500 m will have at most 3 potholes?

Problem 3

A fisherman catches a fish at an average rate of 1.8 per hour. Is he being very ambitious in hoping he will catch 4 or more fish in a 2-hour session?

Problem 4

Blemishes on the paint work of the bonnet of a newly sprayed car occur at random with an average blemish rate of 0.03 per m^2 . If a bonnet has an area of $1.6 m^2$ how likely is it that one randomly selected bonnet will be blemish free if the occurrence of the blemishes is random?

A moment's consideration should convince you that it is not possible to apply the binomial distribution to this problem. Recall that the binomial is applicable if we can identify a number n of identical trials with each trial resulting in one of two possible outcomes.

While we can plausibly identify two outcomes (i.e. there either is or is not an arrival/catch/pothole/blemish) there is no obvious candidate for what should constitute a trial. There are in fact an unlimited number of possibilities for a customer to arrive or for a server to crash or for a pothole to appear on a road.

We deal with these problems using the Poisson distribution function and describe them as Poisson processes

We can calculate the probability of observing k events using the probability mass function

$$\Pr(X = k) = \frac{e^{-\lambda} \lambda^k}{k!} \text{ where } \lambda \text{ is the event rate.}$$

In all of the problems above, events occur at a random time or place with the average rate of such events occurring remaining constant.

However, these events occur independently of one another. E.g. the likelihood of a customer arriving in say the next 5 minutes does not depend on what has happened in the past.

Or the likelihood of finding a pothole say in the next 100m stretch of road does not depend on how many potholes have been encountered up to that point.

The memoryless property

The independence property of a Poisson process is sometimes called the memoryless property. I.e. the past is forgotten and so cannot possibly have an influence in what many happen in the future.

A fisherman who has not caught a fish in 90 minutes may reason to himself (erroneously!) that he is “due” to land one. Of course, the fish don't (and can't!) know that he has had a “bad” 90 minutes and will not present themselves for capture to help redress this.

Rates and inter event times

In any problem describing events that occur according to a Poisson distribution it is necessary to know the frequency at which these events occur. For example, in the queuing problem we are told that customers arrive at an average rate of 1.8 arrivals per minute. We then scaled this up to 3.75 arrivals per $2\frac{1}{2}$ minutes as the question we sought to answer was concerned with the number of arrivals in $2\frac{1}{2}$ minutes (specifically we sought the probability that there would be one arrival).

Or in the fishing problem we were given that fish are caught at a rate of 1.8 per hour which again we scaled up to a rate per 2 hours. In both problems the frequency of the events occurring were given as rates. I.e. so many events per minute or per hour.

In some problems we may be given as an alternative, the average time between events. E.g. a customer arrives on average every 12 minutes or the mean time between failures (MTBF) for a machine is 240 operational hours. You must convert inter event times to event rates before applying the Poisson formula.

Sometimes the conversion is easy. For example, 15 minutes between events on average is clearly equivalent to 4 events per hour as there are 4 intervals of length 15 minutes in an hour.

Or 2 months between events is equivalent to 6 events per year as there are 6 intervals of 2 months in a year.

For a less convenient value such as say 18 minutes between arrivals, it is perhaps best to follow the following procedure.

1. Write down the frequency information exactly as it is given
2. If the information describes an inter event time rather than an event rate simply invert to convert!
3. Scale the rate up to the required time units changing time units (e.g. min $\rightarrow$ hours) if required.

Example

Convert an average customer inter arrival time of 18 minutes to an arrivals per hour rate.

1. 18 minutes / arrival (This has form TIME / EVENT so step 2 is needed)
2. $\equiv \frac{1}{18}$ arrivals / minutes (This is now a rate i.e. EVENTS / TIME as required)
3. $\equiv \frac{60}{18} = 3.33$ arrivals / hour (x by 60 to get from a per minute rate to a per hour rate)

Problem 1

How likely is it that one customer will arrive in the next $2\frac{1}{2}$ minutes?

Problem 2

At most 3 potholes in 1.5 km?

Problem 3

4 or more fish in two hours?

Problem 4

Blemish free?

The (negative) exponential distribution

Sometimes we may wish to consider, not the probability that a particular number of events occur in a given interval of time but the probability that a particular interval of time will elapse before the occurrence of the next event. For example, the manner in which cars stop for a hitchhiker requiring a lift is random in time and so is a plausible candidate for the Poisson model.

However rather than calculating the probability of him getting say at least 2 lifts in the next hour it is more useful to calculate the likelihood of say an $1\frac{1}{2}$ hours passing without him getting a lift.

Such a question is most naturally answered using the negative exponential distribution which can be viewed as a special application of the Poisson distribution.

For example if we know that on average 2 cars stop every hour then we let Y = the number of cars that stop in $1\frac{1}{2}$ hours so $Y \sim \text{Poi}(\lambda)$ with $\lambda = 3$ (Exercise) and the probability then of the hitchhiker getting a lift in $1\frac{1}{2}$ hours = $1 - \Pr(Y = 0)$ which is $\approx 95\%$ (Exercise).

Catastrophic failure

One category of examples where we are more interested in the first occurrence of an event is concerned with the failure of machines or devices. If the Poisson model is to be applicable, then the memoryless property must be evident. I.e. the prognosis for such a device must be no way be influenced by anything in its past such as for example a previous history of failure or perhaps an exemplary service history.

This immediately rules out entire categories of devices such as mechanical devices like cars or organic entities like people or animals.

In fact, the Poisson may only be applied to devices which can only exist in two states. They may be either perfect or destroyed and the transition from the former state to the latter happens instantaneously. Such failure is described as catastrophic failure.

For example, a piece of crockery such a cup or plate has this property. They fail when they break and thus go from perfect working order to destroyed in an instant.

(You may have noticed that the crockery in your cupboard at home do not break in the order in which they were purchased. A plate in use for 20 years is as likely to survive for another year as a newly purchased plate.)

Other devices which are prone to catastrophic failure or approximately so are electrical light bulbs and fuses, web or file servers, many electronic and some electrical devices (which uses the concept of mean time between failures to quantify their reliability), military entities (tanks, fighter planes etc.) in a battle, shear bolts in agricultural machinery etc.

Model fitting and scope

The various examples considered above should give a feel for the kind of stochastic processes which might plausibly be described by a Poisson distribution. Loosely speaking if events happen randomly (e.g. in time) and the underlying rate of occurrence of such events remains the same then a Poisson model may be used.

Of course, as with all mathematical and statistical models, the Poisson model is generally a simplified interpretation of that which it represents and thus any results obtained are only approximations.

For example, the manner in which customers arrive to join a queue is often modelled using a Poisson distribution. In practice many queues have busy periods and quiet periods and so the underlying customer arrival rate is not constant.

Also, people joining queues often do so in the company of family or friends and so such arrival events are not independent of one another. However, if we restrict our application of the model to scenarios where the arrival rate does not vary very much (i.e. limit the scope) and if the number of non-independent arrivals is small then the approximate results may be sufficiently accurate to be useful.

As another example, electrical engineers would dispute that light bulbs fail catastrophically. They would argue that the filament degrades over time and also the gas within a bulb gradually escapes both contributing to a diminishing of the bulb's integrity over time. However empirical evidence (historical data) suggests that the Poisson model fits reasonably well in practice.

In summary it is not appropriate to definitively proclaim that a particular model is or is not appropriate for a particular class of examples. Sometimes the "obvious" choice of model does not fit very well as the assumptions are excessively violated. Sometimes a model which should not logically be considered may happen to fit the data quite well. A number of statistical techniques exist to test the validity of the assumptions made in the Poisson (and other) models.

Applications of the Poisson model

Below is a list of situations where the Poisson model is frequently applied in a satisfactory manner.

The event of customer arrivals per unit time to join a queue
 The number of defects per unit length / area on an item
 The event of lightning strikes per unit time in an electrical storm
 The occurrence of accidents per unit time on a road / in a factory / amongst a population subgroup
 The number of typing errors on a page of a document draft or bugs in a section of computer code
 The scoring of a goal in a football match

Poisson distribution in R

Recall for the arrival problem we let $X = \#$ arrivals in a 2-minute interval and established that we need to find $\Pr(X = 1)$ where $X \sim \text{Poi}(3.75)$. To do this in R type

```
> dpois(1, 3.75) # Ans 0.08819155
```

For the potholes problem we let $X = \#$ potholes per 1,500 m stretch of road and established that we need to find $\Pr(X \leq 3)$ where $X \sim \text{Poi}(2)$. To do this in R we must find $\Pr(X \leq 3)$ which we do with

```
> ppois(3, 2, lower.tail=TRUE) # Ans 0.8571235
```

For the fisherman problem we let $X = \#$ fish caught in 2 hours and established that we need to find $\Pr(X \geq 4)$ where $X \sim \text{Poi}(3.6)$. To do this in R we must find $\Pr(X > 3)$ which we do by typing

```
> ppois(3, 3.6, lower = FALSE) # Ans 0.4847839
```

✎ Be careful with off by one error here. Note that $\text{ppois}(k, \lambda, \text{lower} = \text{TRUE})$ returns $\Pr(X \leq k)$ while $\text{ppois}(k, \lambda, \text{lower} = \text{FALSE})$ returns $\Pr(X > k)$

Exercise

A hard disk drive claims to have a mean time to failure of 17 months. Assuming failure can reasonably be modelled as a Poisson process, how likely is it that it will survive for:

- (a) 2 years? (about 25%)
- (b) 1 year? (about 50%)
- (c) 17 months? (about 37% - **not** 50%)

What would the MTTF need to be if there is to be a 90% chance of the HDD surviving for a year?